Algorithmic trading Strategies

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Abstract

”Picture trades flashing through markets in an instant, guided by the precision of machines. This report dives into the thrilling world of algorithmic trading, where we harness computers to outsmart traditional methods. Using cutting-edge tools like TradeStation, we craft and test automated strategies, blending technical analysis with advanced algorithms. Backtesting unveils their potential to seize market opportunities, promising efficiency and speed beyond human reach. Yet, challenges like market impact and data pitfalls lurk, demanding vigilance. As we gaze toward a future enriched by artificial intelligence, this journey reveals a new frontier in finance—where innovation meets opportunity, poised to redefine trading for the bold and the curious.”

Contents

1 Introduction

The cryptocurrency market has experienced exponential growth in recent years, with Bitcoin (BTC) and Ethereum (ETH) emerging as the most prominent digital assets. These markets are characterized by high volatility, 24/7 trading, and substantial liquidity, making them an ideal domain for algorithmic trading strategies. The Untrade Hackathon, organized in collaboration with Quant Club, BITS Pilani, provided a platform to explore and develop such strategies using historical data for the BTC/USDT and ETH/USDT trading pairs.

In this hackathon, participants were challenged to create as many logic-based algorithmic trading strategies as possible, aimed at generating profitable trading signals for the specified cryptocurrency markets. Our approach centered on designing and implementing multiple strategies rooted in technical analysis, a widely adopted method for predicting price movements based on historical market data. We utilized several well-known indicators, such as moving averages, the relative strength index (RSI), and the moving average convergence divergence (MACD), to construct our trading logic.

2 Related Work

Algorithmic trading in cryptocurrency markets has been widely studied due to their volatility and liquidity. Early research, explored price manipulation in Bitcoin markets, highlighting the need for robust strategies. Technical analysis, rooted in indicators such as moving averages and RSI, has been a mainstay, with studies validating its use in crypto. Recent advancements incorporate machine learning for predictive modeling ,though often limited by data quality. Unlike these efforts, our hackathon work focuses on diversifying logic-based strategies, leveraging historical data to address gaps in prior adaptability and breadth.

3 Theory

Algorithmic trading refers to the use of computer algorithms to automate the buying and selling of financial instruments based on predefined rules or strategies. In the context of cryptocurrency markets, such as BTC/USDT and ETH/USDT, algorithmic trading leverages the high volatility and continuous trading hours to identify profitable opportunities.

A cornerstone of many algorithmic trading strategies is technical analysis, which involves studying historical price and volume data to identify patterns and trends that can forecast future price movements. Unlike fundamental analysis, which assesses an asset's intrinsic value based on economic and financial factors, technical analysis relies solely on market-generated data.

In this section, we outline the technical indicators that form the foundation of our logic-based trading strategies.

4 Strategy 1 (BTC/USDT)

4.1 Average True Range

ATR (Average True Range): A technical indicator that measures market volatility by calculating the average range of price movements over a specified period n.

$$ATR = \frac{1}{n}\sum_{i=1}^{n} TR_i, \quad \text{where} \quad TR_i = \max\left(\text{high}_i - \text{low}_i, \left|\text{high}_i - \text{close}_{i-1}\right|, \left|\text{low}_i - \text{close}_{i-1}\right|\right)$$

· n (Lookback Period): The number of periods (e.g., days) over which the ATR is calculated.

· The difference between the high and low prices of the current period: $(\text{high}_i - \text{low}_i)$

· The absolute difference between the current high price and the previous close price: $\left|\text{high}_i - \text{close}_{i-1}\right|$

4.2 Relative Strength Index (RSI)

The Relative Strength Index (RSI) is a momentum oscillator that quantifies the speed and magnitude of price changes. It is calculated as:

$$\text{RSI} = 100 - \frac{100}{1 + \frac{\text{average gain}}{\text{average loss}}}$$

where the average gain and loss are typically computed over 14 periods. RSI values range from 0 to 100, with levels above 70 indicating overbought conditions (potential sell signal) and below 30 indicating oversold conditions (potential buy signal).

4.3 Moving Average Convergence Divergence (MACD)

The Moving Average Convergence Divergence (MACD) is a trend-following indicator that captures the relationship between two EMAs of a security's price. It comprises:

- MACD line: the difference between the 12-period EMA and the 26-period EMA.
- Signal line: a 9-period EMA of the MACD line.
- Histogram: the difference between the MACD line and the signal line.

A buy signal occurs when the MACD line crosses above the signal line, while a sell signal occurs when it crosses below.

4.4 Average Directional Index(ADX)

The Average Directional Index (ADX): It is a technical indicator that measures the strength of a trend in a financial market. It is derived from the +DI (Positive Directional Indicator) and -DI (Negative Directional Indicator), which represent the strength of upward and downward movements, respectively.

$$ADX = \frac{1}{n}\sum_{i=1}^{n} \frac{\left|(+DI_i - (-DI_i))\right|}{(+DI_i + (-DI_i))}$$

4.5 Random Forest

Complex Pattern Recognition:
Reason: Financial markets are noisy and non-linear. Random Forest excels at identifying intricate relationships between technical indicators (e.g., rsi, macd, adx) and future price movements, beyond simple rule-based strategies.
Benefit: It learns from multiple features simultaneously, improving prediction accuracy over manual thresholds.
It's implemented by preparing feature and target data, training a 150-tree classifier, predicting signals across all periods, and translating those signals into actionable trades via a simple loop.

Algorithm Overview

4.6 Inputs and Outputs

- **Input:** DataFrame with OHLCV data, initial balance (default: 10,000)
- **Output:** DataFrame with trading signals and position sizing

4.7 Trading Strategy Algorithm

Algorithm 1: Strategy 1

```
Input: DataFrame data with columns: datetime, open, high, low, close, volume;
       initial_balance=10000
Output: DataFrame with trade_type, leverage, position

if data invalid then
    Log error; Assign HOLD, leverage=1, position=50;
    return data;
end
Set df = data copy; Index by datetime;
Compute indicators: ATR(14), EMA(21), RSI(14), MACD, ADX(14), volume_ratio, volatility,
  sharpe;
Fill NaNs with 0;
Set target = 1 if future_return >0.0025 and RSI >50, -1 if <-0.0002 and RSI <50, else 0;
Train RandomForest(features, target, params: n=150, depth=5);
Predict ml_signal for all rows;

Initialize trade_type = HOLD, position_open = False;
for each row i in df do
    if not position_open then
        if ml_signal[i] == 1 then
            trade_type[i] = LONG; position_open = True;
        else
            if ml_signal[i] == -1 then
                trade_type[i] = SHORT; position_open = True;
            end
        end
    end
    else
        trade_type[i] = CLOSE; position_open = False;
    end
end
Set leverage = 1 if ATR >75th percentile, else 2;
Set position = clip(0.003 * initial_balance / ATR, 50, 100);
return df with trade columns;
```

4.8 Main Execution

- Generate sample data with random prices
- Create Strategy instance
- Run strategy on sample data
- Print results

4.9 Output

Metric	Value
textAverage Drawdown	2.03
Average TTR	18.33 days
Final Balance	8,827.71
Initial Balance	1,000.00
Leverage	1.00
Max Portfolio Balance	8,827.71
Maximum Drawdown	17.43
Maximum PNL	403.76
Minimum PNL	-373.80
Minimum Portfolio Balance	1,000.00
Number of Trades	205
Profit (%)	782.77
TTR (Total Time to Recover)	79 days
Total Fee	1,019.92
Flag	1

Table 1: Compounding Strategy

Metric	Value
Avg. Adverse Excursion	1.71
Avg. Drawdown (%)	0.92
Avg. Holding Time	1 day
Avg. Loss	-20.01
Avg. Profit	11.00
Avg. TTR	18.33 days
Avg. Win	19.23
Benchmark Return (%)	560.46
Benchmark Return ($1000)	5,604.62
From	2020-07-01
Gross Profit	2,515.46
Largest Loss	-98.77
Largest Win	87.21
Losing Streak	3
Losing Trades	43
Max. Adverse Excursion	14.05
Max. Drawdown (%)	9.95
Max. Holding Time	1 day
Net Profit	2,255.21
No. of Long Trades	87
No. of Short Trades	118
Sharpe Ratio	9.96
Sortino Ratio	19.69
To	2024-06-29
Total Trades	205
Win Rate (%)	79.02
Winning Streak	23
Winning Trades	162

Table 2: Static Strategy

4.10 Quarterly Results

Initial Balance	Final Balance	Profit (%)	Benchmark (%)	Beaten	From	To	Total Trades	Long Trades	Short Trades	Win Rate (%)
1000.00	1091.94	9.19	16.68	No	2020-07-01	2020-09-30	8	3	5	100.00
1000.00	1217.57	21.76	172.80	No	2020-10-01	2020-12-31	17	17	0	82.35
1000.00	1304.30	30.43	100.45	No	2021-01-01	2021-03-31	18	16	2	72.22
1000.00	972.80	-2.72	-40.42	Yes	2021-04-01	2021-06-30	13	1	12	61.54
1000.00	1348.37	34.84	30.78	Yes	2021-07-01	2021-09-30	14	6	8	85.71
1000.00	1241.85	24.18	-3.98	Yes	2021-10-01	2021-12-31	11	7	4	90.91
1000.00	1055.67	5.57	-4.61	Yes	2022-01-01	2022-03-31	12	3	9	83.33
1000.00	1232.99	23.30	-56.94	Yes	2022-04-01	2022-06-30	13	1	12	84.62
1000.00	1033.99	3.40	0.72	Yes	2022-07-01	2022-09-30	9	0	9	55.56
1000.00	942.75	-5.73	-14.33	Yes	2022-10-01	2022-12-31	17	0	17	64.71
1000.00	1271.47	27.15	71.31	No	2023-01-01	2023-03-31	15	12	3	93.33
1000.00	1073.29	7.33	7.09	Yes	2023-04-01	2023-06-30	9	1	8	66.67
1000.00	1039.45	3.95	-11.85	Yes	2023-07-01	2023-09-30	12	0	12	50.00
1000.00	1149.56	14.96	51.22	No	2023-10-01	2023-12-31	11	10	1	90.91
1000.00	1184.59	18.46	61.34	No	2024-01-01	2024-03-31	15	10	5	100.00
1000.00	1081.81	8.18	-9.33	Yes	2024-04-01	2024-06-23	8	1	7	75.00

Table 3: Quarterly Trading Performance

4.11 Results

In the last 16 quarters:

- 14 quarters are in profit.
- Average win rate is close to 60 percent in every quarter.

5 Strategy 2 (ETH/USDT)

5.1 Exponential Moving Average (EMA)

A type of moving average that gives more weight to recent prices, making it more responsive to new data compared to a simple moving average. It's widely used to identify trends and smooth price fluctuations in trading strategies.

$$EMA_t = (\text{close}_t \times \alpha) + (EMA_{t-1} \times (1 - \alpha))$$
$$\alpha = \frac{2}{\text{window} + 1}$$

5.2 On-Balance Volume (OBV) and OBV Oscillator

OBV (On-Balance Volume): Tracks cumulative volume flow to confirm price trends. Rising OBV with rising prices suggests buying pressure; falling OBV with falling prices suggests selling pressure.
OBV Oscillator: Measures momentum of OBV by comparing it to its short-term EMA, highlighting overbought/oversold conditions or trend shifts.

5.3 Moving Average Convergence Divergence (MACD)

A trend-following momentum indicator that shows the relationship between two exponential moving averages (EMAs) of a security's price. It helps identify potential buy/sell signals and trend strength/direction. Positive values of MACD suggest bullish momentum, while negative values indicate bearish momentum, aiding trade signal generation.

5.4 Relative Strength Index (RSI)

A momentum oscillator that measures the speed and change of price movements to identify overbought or oversold conditions. It ranges from 0 to 100 and is used to assess potential reversals or trend strength.

$$RSI = 100 - \left(\frac{100}{1 + RS}\right)$$
$$RS = \frac{\text{Average Gain}}{\text{Average Loss}}$$
$$\text{Average Gain} = \text{Rolling average of upward price changes over the window}$$
$$\text{Average Loss} = \text{Rolling average of downward price changes over the window (absolute values)}$$

5.5 Average Directional Index (ADX)

A technical indicator that measures the strength of a price trend (not its direction) on a scale of 0 to 100.
see: sec 4.4 for more detail

Algorithm 2: ETH Trading Strategy Pseudocode

```
Input: DataFrame data with columns: datetime, open, high, low, close, volume;
       initial_balance=10000
Output: DataFrame with trade_type, leverage, position

if data invalid then
    Log error; Assign HOLD, leverage = 1, position = 50;
    return data;

Set df = data copy; Index by datetime;
Rename 'volume from' to 'volume' if present;
Compute indicators: ATR(14), EMA(21), RSI(14), MACD, ADX(14), volume_ratio, volatility,
  sharpe, OBV, OBV_osc; Fill NaNs with 0;
Set target = 1 if future_return >0.0015 and RSI >50, -1 if <-0.0015 and RSI <50, else 0;
Train RandomForest(features, params: n=150, depth=5); Predict ml_signal;

// Trading Logic
Initialize trade_type = HOLD, position_open = False;
Compute atr_quantile = atr.rolling(20).quantile(0.9);
for each row i in df do
    Set signal = ml_signal[i], price = close[i], date = index[i];
    if atr[i] >atr_quantile[i] then
        Continue;
    if position_open then
        if (date - entry_date).days >= 5 then
            trade_type[i] = CLOSE; Reset vars; Continue;
        if position_type = LONG then
            Update trailing_sl = max(trailing_sl, price * 0.999);
            if price <= sl or price >= tp or price <= trailing_sl then
                trade_type[i] = CLOSE; Reset vars;
        else
            Update trailing_sl = min(trailing_sl, price * 1.001);
            if price >= sl or price <= tp or price >= trailing_sl then
                trade_type[i] = CLOSE; Reset vars;
    else
        if signal == 1 then
            trade_type[i] = LONG; position_open = True; position_type = LONG;
            Set sl = price * 0.997, tp = price * 1.002, trailing_sl = price * 0.999, entry_date =
              date;
        else
            if signal == -1 then
                trade_type[i] = SHORT; position_open = True; position_type = SHORT;
                Set sl = price * 1.003, tp = price * 0.998, trailing_sl = price * 1.001, entry_date
                  = date;
            else
                if signal == 0 then
                    trade_type[i] = CLOSE; Reset vars;

Set leverage = 1, position = clip(0.00003 * initial_balance / atr, 50, 100);
return df with trade columns;
```

5.6 Output

Metric	Value
Average Drawdown	0.78
Average TTR	13.59 days
Final Balance	7,497.473
Initial Balance	1,000.00
Leverage	1.00
Max Portfolio Balance	7532.31
Maximum Drawdown	7.30
Maximum PNL	321.02
Minimum PNL	-178.57
Minimum Portfolio Balance	1,000.00
Number of Trades	241
Profit (%)	649.747
TTR (Total Time to Recover)	77 days
Total Fee	633.47
Flag	1

Table 4: Compounding Strategy

Metric	Value
Avg. Adverse Excursion	1.99
Avg. Drawdown (%)	0.43
Avg. Holding Time	1 days 0:23:54
Avg. Loss	-13.066
Avg. Profit	8.53
Avg. TTR	13.59 days
Avg. Win	14.18
Benchmark Return (%)	1,361.98
Benchmark Return ($1000)	13,619.83
From	2020-07-01
Gross Profit	2,239.00
Largest Loss	-44.26
Largest Win	80.98
Losing Streak	4
Losing Trades	50
Max. Adverse Excursion	14.19
Max. Drawdown (%)	5.005
Max. Holding Time	2 day
Net Profit	2,055.25
No. of Long Trades	173
No. of Short Trades	68
Sharpe Ratio	10.68
Sortino Ratio	23.57
To	2024-06-29
Total Trades	241
Win Rate (%)	79.25
Winning Streak	17
Winning Trades	191

Table 5: Static Strategy

5.7 Quarterly Results

Initial Balance	Final Balance	Profit (%)	Benchmark (%)	From	To	Total Trades	Long Trades	Short Trades	Win Rate (%)
1000.00	1107.73	10.77	55.70	2020-07-01, 00:00	2020-09-30, 23:59	12	8	4	91.67
1000.00	1160.35	16.04	109.11	2020-10-01, 00:00	2020-12-31, 23:59	23	23	0	78.26
1000.00	1299.17	29.92	163.52	2021-01-01, 00:00	2021-03-31, 23:59	23	19	4	78.26
1000.00	1219.76	21.98	15.50	2021-04-01, 00:00	2021-06-30, 23:59	16	10	6	81.25
1000.00	1267.59	26.76	42.48	2021-07-01, 00:00	2021-09-30, 23:59	14	12	2	92.86
1000.00	1050.54	5.05	11.11	2021-10-01, 00:00	2021-12-31, 23:59	10	9	1	60.00
1000.00	1118.50	11.85	-12.80	2022-01-01, 00:00	2022-03-31, 23:59	19	11	8	78.95
1000.00	1117.54	11.75	-69.00	2022-04-01, 00:00	2022-06-30, 23:59	14	3	11	78.57
1000.00	1151.03	15.10	25.35	2022-07-01, 00:00	2022-09-30, 23:59	16	10	6	68.75
1000.00	1118.15	11.81	-8.79	2022-10-01, 00:00	2022-12-31, 23:59	17	5	12	76.47
1000.00	1193.06	19.31	51.74	2023-01-01, 00:00	2023-03-31, 23:59	12	11	1	91.67
1000.00	1051.19	5.12	6.24	2023-04-01, 00:00	2023-06-30, 23:59	10	10	0	80.00
1000.00	1005.61	0.56	-13.20	2023-07-01, 00:00	2023-09-30, 23:59	15	4	11	80.00
1000.00	1149.89	14.99	31.76	2023-10-01, 00:00	2023-12-31, 23:59	18	18	0	72.22
1000.00	1128.73	12.87	54.96	2024-01-01, 00:00	2024-03-31, 23:59	13	13	0	76.92
1000.00	1049.68	4.97	8.74	2024-04-01, 00:00	2024-06-04, 00:00	9	7	2	88.89

Table 6: Quarterly Trading Performance

5.8 Results

In the last 16 quarters:

- None of the quarters are in loss (without overfitting).
- Profitable in 16 out of 16 quarters.
- Average win rate is close to 75 percent in every quarter.